The Student-t Multi-Fidelity Objective

`zrg_cost_function_student_t`

The full, explicit scalar function cost_fn(x) that L-BFGS-B minimizes over the 19 free parameters x in the MF Student-t runs (e.g. aurora_mf_ne128_ne32_prod_studentt_1000starts_annual.yaml), built up step by step, with this run's actual configuration (dof=4, kappa=0, lambda=0, and the PCP/OSR/OLR/TLWP weights) plugged in throughout.

What v_c actually is, before anything else

It is the GP's own posterior predictive variance at the candidate point x, for one specific (location, variable) pair -- m, v = gp.predict(x), then v_c = max(v, var_floor). It is NOT the GP kernel's variance hyperparameter (hyperparameters.pkl's k0_variance/ k1_variance, a single fixed number set once during training) and NOT a model of real observational/physical noise (there is no separate var_obs term anywhere in this cost). It is specifically the GP's own uncertainty about its own prediction at this particular x -- small near training data, large in sparse regions -- which is exactly why moving into a sparse region can inflate v_c, and is the mechanism the whole κ/boundary-pinning discussion is about. Mathematically it plays the role of σ² in the density on the next page; its source is GP predictive uncertainty at a point, not a fixed hyperparameter or a noise model.

Step 0 — the actual starting point: the Student-t density

Everything below is -log of one specific, named probability density — it doesn't start from an already-decomposed cost formula, it starts from this. For residual r, scale v_c, and ν degrees of freedom:

$$p(r \mid v_c, \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi v_c}} \left(1 + \frac{r^2}{\nu v_c}\right)^{-\frac{\nu+1}{2}}$$

$\ell(r, v) := -\log p(r \mid v_c, \nu)$ is, by definition, the per-location cost term. Expanding -log of the density directly (log of a product is a sum of logs; log of a power pulls the exponent out front):

$$-\log p = -\ln\Gamma\left(\tfrac{\nu+1}{2}\right) + \ln\Gamma\left(\tfrac{\nu}{2}\right) + \tfrac{1}{2}\ln(\nu\pi) + \tfrac{1}{2}\log v_c + \tfrac{\nu+1}{2}\log\left(1 + \tfrac{r^2}{\nu v_c}\right)$$

$$= \text{const}(\nu) \; + \; \tfrac{1}{2}\log v_c \; + \; \tfrac{\nu+1}{2}\log\left(1 + \tfrac{r^2}{\nu v_c}\right)$$

which is exactly Step 1's l(r,v) below (minus the κv_c term — that is not part of this density, it's an extra regularization term added on top afterward, same role κ plays in the Gaussian-NLL cost, and the reason κ>0 makes this a log-posterior rather than a pure likelihood).

What prior does $\nu = 4$ actually put on the variance itself?

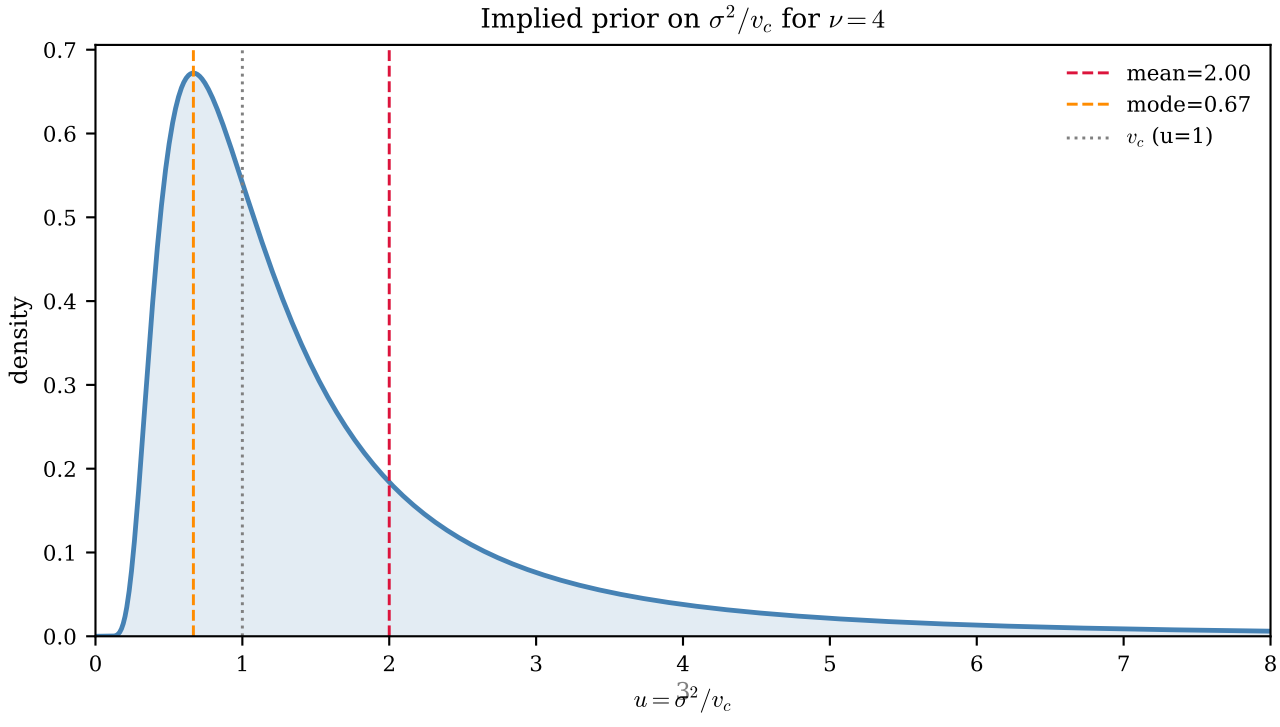
Student-t was motivated as "Gaussian with the variance itself uncertain" — a Normal with unknown variance, given an Inverse-Gamma prior on that variance and marginalized out, is exactly a Student-t. Concretely, if

$$r \mid \sigma^2 \sim N(0, \sigma^2), \qquad \sigma^2 \sim \text{InvGamma}\left(\tfrac{\nu}{2}, \tfrac{\nu v_c}{2}\right)$$

then marginalizing σ^2 out reproduces $p(r \mid v_c, \nu)$ from Step 0 exactly — verified numerically (marginalizing the Gaussian over this Inverse-Gamma prior by quadrature vs. evaluating Step 0's density directly matches to ~1e-7, i.e. numerical-integration noise, at several values of r).

For this run's $\nu = 4$: shape=2, scale=$2v_c$

In units of $u = \sigma^2 / v_c$ (so the plot below is universal, not tied to any specific run's v_c), that's $u \sim \text{InvGamma}(2, 2)$, with mean 2 and mode 0.67 — the implied belief is that the true variance is typically a couple of multiples of the GP's own v_c, not equal to it, and noticeably right-skewed. One more fact worth knowing: the variance of this variance prior itself is undefined (infinite) at exactly $\nu = 4$ — $\text{Var}[\text{InvGamma}(\alpha, \beta)]$ needs $\alpha > 2$, and here $\alpha = \nu/2 = 2$ sits right at that boundary: Student-t's kurtosis is finite only for $\nu > 4$, so $\nu = 4$ is exactly the edge of how heavy a tail you can have while the tail's own uncertainty still has a well-defined second moment.



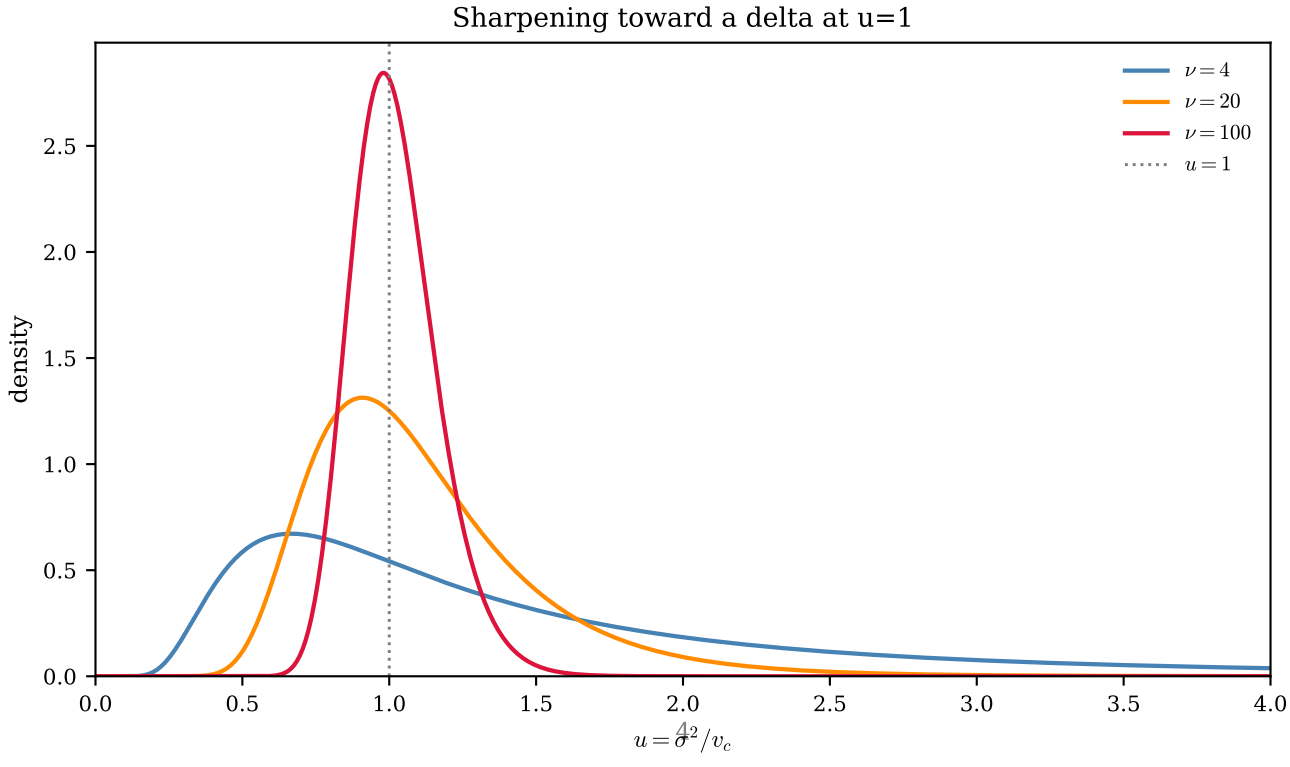
One prior per call, and what happens as $\nu \to \infty$

This prior is not global — it's a separate InvGamma(ν/2, νv_c/2) implicitly assumed at every individual (location, variable) evaluation of l(r,v), since v_c is different at every one of the 88 cells (and changes again every time the optimizer evaluates a new x, since v_c comes fresh from gp.predict(x) each time). Only the shape (α=ν/2) is shared globally; the scale tracks whatever v_c the GP outputs at that specific cell.

As $\nu \to \infty$, this per-call prior collapses to a delta function at $u = 1$ — i.e. "the true variance *is* v_c, with no uncertainty", which is exactly the assumption a plain Gaussian makes. Mean and mode both converge to 1, and the spread shrinks to 0:

$\nu = 4$	mean=2.000	mode=0.667	std=∞ (undefined below ν=4)
$\nu = 20$	mean=1.111	mode=0.909	std=0.393
$\nu = 100$	mean=1.020	mode=0.980	std=0.147
$\nu = 1{,}000$	mean=1.002	mode=0.998	std=0.045
$\nu = 10^5$	mean=1.00002	mode=0.99998	std=0.004

($\text{mean}(u) = \frac{\nu}{\nu-2} \to 1$, $\text{mode}(u) = \frac{\nu}{\nu+2} \to 1$, $\text{std}(u)$ requires $\nu > 4$ to exist at all and shrinks like $\sim \sqrt{2/\nu} \to 0$ — the same $\nu \to \infty$ limit proved algebraically for the cost term a few pages on, viewed here from the prior side instead of the NLL-formula side.)



Integrating over the variance vs. taking $\nu \to \infty$

It's tempting to think "Student-t comes from marginalizing something out, so maybe marginalizing ν itself gets back to Gaussian." It doesn't — these are different kinds of operation:

- Integrating over σ^2, at fixed ν, turns a Gaussian into a Student-t (the derivation above). ν is not integrated over here — it's the fixed hyperparameter shaping the prior being integrated.
- Taking the limit $\nu \to \infty$ turns Student-t back into a Gaussian (next page). This is a limit, not an integral — there is no p(ν) being summed against; it's one sequence converging to an endpoint.

Integrating out a prior $p(\nu)$ instead:

$$\int \text{Student-}t(r;\nu)\, p(\nu)\, d\nu$$

would generally NOT give a Gaussian back: every finite-ν Student-t already has heavier-than-Gaussian tails, and mixing distributions together can only preserve or worsen tail weight, never lighten it. The only p(ν) for which this collapses to a Gaussian is a point mass at ν=∞ — not really an integration, just the same limit again.

The Bayesian hierarchical model, written out fully

Step 0's marginalization in standard likelihood / prior / marginal notation, for one (location, variable) call, ν fixed:

$$r \mid \sigma^2 \sim \text{Normal}(0, \sigma^2)$$

$$\sigma^2 \sim \text{InvGamma}(\alpha = \nu/2,\ \beta = \nu v_c/2)$$

which marginalizes (Step 0) to $r \sim \text{Student-}t(\nu)$:

$$p(r \mid v_c, \nu) = \int N(r \mid 0, \sigma^2)\ \text{InvGamma}(\sigma^2 \mid \nu/2,\ \nu v_c/2)\ d\sigma^2$$

v_c is the prior's central reference point: $u = 1$ ($\sigma^2 = v_c$) is where "true variance equals v_c" sits. The actual mode is slightly below that, at $\frac{\nu}{\nu+2} v_c$ (recall $\nu = 4$: mode $= 0.667\, v_c$, not v_c) — converging to v_c only as $\nu \to \infty$. ν controls how tightly that's believed — large ν = "confident v_c is right" (plain Gaussian in the limit), small ν = "not confident, true variance could differ a lot." The optimizer never sees or estimates σ² itself — only the already-marginalized r, with the variance-uncertainty baked in.

Step 1 — the per-location Student-t NLL term

Restating the result of Step 0 with the optional variance-penalty term included:

$$\ell(r, v) = \tfrac{1}{2}\log v_c \;+\; \tfrac{\nu+1}{2}\log\!\left(1 + \tfrac{r^2}{\nu v_c}\right) \;+\; \text{const}(\nu) \;+\; \kappa v_c$$

where $v_c = \max(v,\, 10^{-6})$, and

$$\text{const}(\nu) = \;-\ln\Gamma\!\left(\tfrac{\nu+1}{2}\right) + \ln\Gamma\!\left(\tfrac{\nu}{2}\right) + \tfrac{1}{2}\ln(\nu\pi)$$

is the Student-t normalizing constant — fixed once ν is chosen, independent of r or v. It does not move the argmin, but makes the returned value a true negative log-likelihood.

This run's numbers: $\nu = 4,\ \kappa = 0$

The κ·v_c term vanishes. const(4) = −lgamma(2.5) + lgamma(2) + 0.5·log(4π) ≈ −0.2847 + 0 + 1.2655 ≈ 0.9808, so:

$$\ell(r, v) = \tfrac{1}{2}\log v_c + 2.5\log\!\left(1 + \tfrac{r^2}{4\,v_c}\right) + 0.9808$$

As $\nu \to \infty$ this converges exactly to the Gaussian NLL term $\tfrac{1}{2}\tfrac{r^2}{v_c} + \tfrac{1}{2}\log(2\pi v_c)$ — Student-t is the same Gaussian-variance idea with heavier tails, recovering Gaussian exactly in the limit.

Why the $\nu \to \infty$ limit is exact

The classical fact this is proving: the Student-t density itself converges pointwise to the Gaussian density as $\nu \to \infty$ (Step 0). Since $-\log$ is continuous wherever a density is positive, that carries straight through to $\ell(r, v) = -\log p(r \mid v_c, \nu)$ — the algebra below just makes that carry-through explicit, term by term, rather than asking you to take the density-level fact on faith. The two ν-dependent pieces (the fit term and const(ν)) converge separately.

The fit term

Let $x = r^2/(\nu v_c) \to 0$ as $\nu \to \infty$. Using $\log(1+x) = x - x^2/2 + O(x^3)$:

$$0.5(\nu+1)\log\left(1 + \tfrac{r^2}{\nu v_c}\right) = 0.5\left(1+\tfrac{1}{\nu}\right)\tfrac{r^2}{v_c} - O\left(\tfrac{1}{\nu}\right) \;\to\; \tfrac{1}{2}\tfrac{r^2}{v_c}$$

the 1/ν correction vanishing, leaving exactly the Gaussian squared-error term.

The normalizing constant

Less obvious: three pieces individually blow up or vanish, and the claim is they cancel to one finite number. Using Stirling's series $\ln\Gamma(z) = (z-\tfrac{1}{2})\ln z - z + \tfrac{1}{2}\ln(2\pi) + O(1/z)$, with $x = \nu/2$:

$$\ln\Gamma(x) - \ln\Gamma(x+\tfrac{1}{2}) = -\tfrac{1}{2}\ln x + O(1/x)$$

so, with $x = \nu/2$:
$-\ln\Gamma(\tfrac{\nu+1}{2}) + \ln\Gamma(\tfrac{\nu}{2}) = -\tfrac{1}{2}\ln\nu + \tfrac{1}{2}\ln 2 + O(1/\nu)$

Adding the third piece, $\tfrac{1}{2}\ln(\nu\pi) = \tfrac{1}{2}\ln\nu + \tfrac{1}{2}\ln\pi$, the $\ln\nu$ terms cancel exactly:

$$\text{const}(\nu) = \tfrac{1}{2}\ln 2 + \tfrac{1}{2}\ln\pi + O(1/\nu) = \tfrac{1}{2}\ln(2\pi) + O(1/\nu)$$

exactly the Gaussian normalizing constant — the two log(ν) divergences cancel each other rather than either side blowing up.

Numerically confirming the const(ν) limit

Not just asymptotically — this is where the O(1/ν) error term actually lands:

$\nu = 4$	const=0.980829	0.5·ln(2π)=0.918939	diff=6.19e-02
$\nu = 100$	const=0.921438	0.5·ln(2π)=0.918939	diff=2.50e-03
$\nu = 10{,}000$	const=0.918964	0.5·ln(2π)=0.918939	diff=2.50e-05
$\nu = 10^6$	const=0.918939	0.5·ln(2π)=0.918939	diff=2.51e-07

diff shrinking ~100x every time ν grows 100x confirms the O(1/ν) error term directly.

Step 2 — mean over locations, within one block, one variable

l is computed at every (zonal band, region, global point) location. Within one ZRG block (zonal, or regional) and one variable, the per-location values are averaged — a plain mean, since zonal_weights and regional_weights are both null in this config. The global block has a single location, so no averaging is needed — it's just l evaluated once.

Zonal and regional locations are never pooled into the same mean — they are sliced apart before any averaging happens (feature layout: [zonal_0 ... zonal_14, region_0 ... region_5, global]).

Step 3 — weighted sum over the 4 variables

At each block level, the per-variable means are combined via a weighted sum:

$$\text{zonal_block} = \sum_v w_v \cdot \overline{\ell}_{\text{zonal}}(v)$$

$$\text{regional_block} = \sum_v w_v \cdot \overline{\ell}_{\text{regional}}(v)$$

$$\text{global_block} = \sum_v w_v \cdot \ell_{\text{global}}(v)$$

This run's variable weights:

PCP	0.3	
OSR	0.1	(downweighted)
OLR	0.3	
TLWP	0.3	

Step 4 — weighted sum over zonal / regional / global

$$\text{snap_cost} = \tfrac{1}{3}\,\text{zonal_block} + \tfrac{1}{3}\,\text{regional_block} + \tfrac{1}{3}\,\text{global_block}$$

A fixed 3-way split, not proportional to location count — without it the 15 zonal bands would numerically dominate the 6 regions and especially the single global point, just by outnumbering them.

Step 5 — weighted sum over snapshots

$$\text{total_cost} = \sum_{\text{snapshots}} dy_w[\text{snap}] \cdot \text{snap_cost}[\text{snap}]$$

One snapshot here (ANN, weight 1.0), so this is a no-op: total_cost = snap_cost.

Step 6 — the training-distance penalty (inert in this run)

The full cost_fn(x) adds one more term on top:

$$\text{cost_fn}(x) = \text{total_cost} + \lambda \cdot \max(0,\ d(x) - \tau)$$

d(x) is x's distance to the nearest real training point (each parameter's own bounds_from_data-rescaled space); τ is the training ensemble's own worst-case internal spacing. This run sets λ = 0, so this term is identically zero for every x — it does not affect this run's optimization, but is structurally part of every cost_fn call.

The full assembled objective

Putting Steps 1–6 together, for this run's configuration:

$$\text{cost_fn}(x) = \tfrac{1}{3}\sum_v w_v \overline{\ell}_{15\,\text{zonal}} + \tfrac{1}{3}\sum_v w_v \overline{\ell}_{6\,\text{regions}} + \tfrac{1}{3}\sum_v w_v\, \ell_{\text{global}}$$

with $\ell(r, v) = \frac{1}{2}\log v_c + 2.5\log(1 + \frac{r^2}{4v_c}) + 0.9808$,
$(m(x), v(x)) = \texttt{gp.predict}(x)$ the MF AR1 GP's HF posterior mean/variance, $r = m(x) - \text{obs}$, $v_c = \max(v(x), 10^{-6})$.

This single scalar is what all 1000 L-BFGS-B starts independently minimize over the 19 free parameters.

This run's configuration, at a glance

Likelihood	student_t
Degrees of freedom ν	4
Variance floor	1e-6
κ (variance penalty)	0 (off)
λ (distance penalty)	0 (off)
n_xstarts	1000
max_workers	192